A Two-Dimensional Counterexample to Convexity Along Real 2-Interpolations

Automatic Math Discovery Run `fa_banach_001`
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Status. Candidate full negative answer, likely valid, under the nontrivial smooth-norm interpretation intended by the source. An explicit C^{ω}_{+} unconditional planar body gives a negative second variation; Cauchy–Kowalevski turns it into an actual local real 2-interpolation family with nonconvex logarithmic partition function.

Abstract

Cordero-Erausquin and Klartag conjectured that $\alpha(t) = -\log \int_{\mathbb{R}^n} e^{-F_t}$ is convex along real 2-interpolation families whose endpoints are even, convex, and 2-homogeneous. We disprove the conjecture already in dimension two. Let K have support function $h(\theta) = 1 - \frac{1}{100}\cos(4\theta)$ and let the homogeneous initial speed have boundary value $a(\theta) = \cos(2\theta)$. Cone coordinates reduce the required second-variation inequality to

$$\int_0^{2\pi} h^2 (a')^2 \, d\theta \geq 4 \int_0^{2\pi} a^2 h(h + h'') \, d\theta.$$

The exact left-minus-right deficit is $8\pi\varepsilon(3 + 4\varepsilon) = -148\pi/625 < 0$ for $\varepsilon = -1/100$. The homogeneous interpolation equation is an analytic noncharacteristic PDE in angular coordinates, so the Cauchy–Kowalevski theorem produces a short analytic 2-interpolation through this datum. After affine time rescaling, its two endpoints are unconditional smooth norm squares on $[0, 1]$, while $\alpha''(1/2) < 0$.

1 Source conjecture

The source is Cordero-Erausquin and Klartag [1], arXiv:1109.3652, published in *Geometric Aspects of Functional Analysis*, Lecture Notes in Mathematics 2050 (2012), 151–168. Conjecture 10 on PDF page 11 states that if the endpoints of a real 2-interpolation are even, convex, and 2-homogeneous norm squares, then the associated function α is convex.

More precisely, the source asks whether, for $F_i(x) = \lambda_i \|x\|^2_{K_i}$ with centrally symmetric convex bodies K_i, "the function α associated with the 2-interpolation family is convex." The source notes that evenness, convexity, and homogeneity propagate along the interpolation.

For a sufficiently regular family $F(t, x)$, real 2-interpolation means

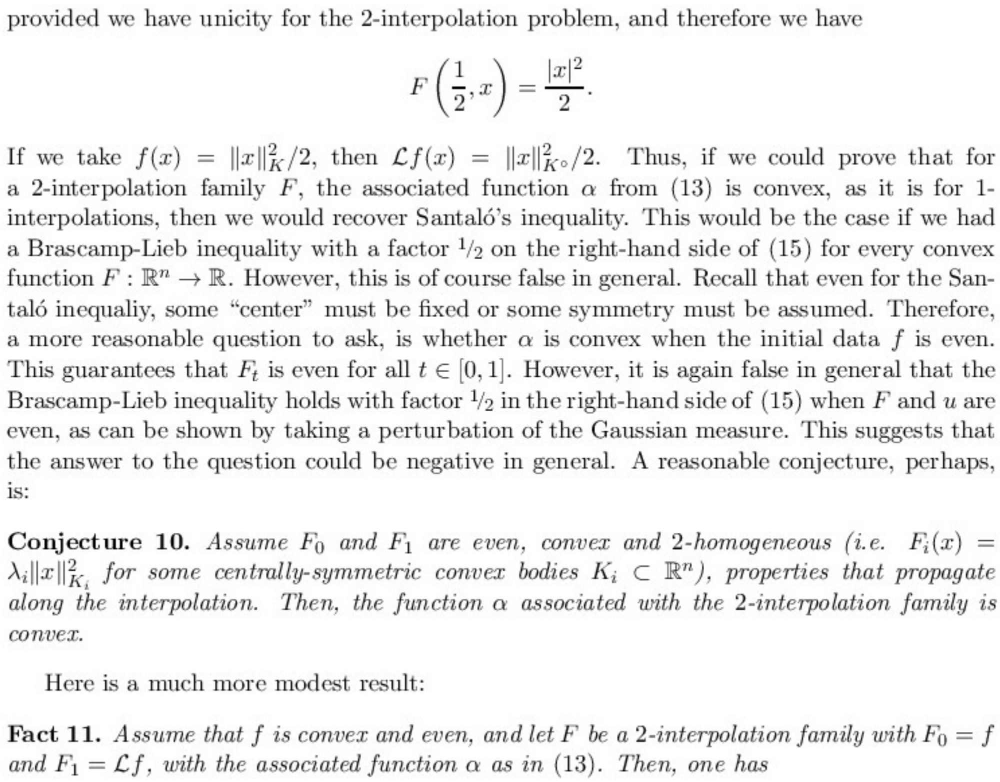
$$\partial_{tt} F = \frac{1}{2} (\operatorname{Hess}_x F)^{-1} \nabla_x \partial_t F \cdot \nabla_x \partial_t F, \tag{1}$$

and

$$\alpha(t) = -\log \int_{\mathbb{R}^2} e^{-F(t,x)} \, dx. \tag{2}$$

Differentiation under the integral gives, with $d\mu_F = e^{-F} dx / \int e^{-F}$,

$$\alpha''(t) = \frac{1}{2} \int (\operatorname{Hess} F)^{-1} \nabla u \cdot \nabla u \, d\mu_F - \operatorname{Var}_{\mu_F}(u), \qquad u = \partial_t F. \tag{3}$$



provided we have unicity for the 2-interpolation problem, and therefore we have

$$F\left(\frac{1}{2}, x\right) = \frac{|x|^2}{2}.$$

If we take $f(x) = \|x\|_K^2/2$, then $\mathcal{L}f(x) = \|x\|_{K^\circ}^2/2$. Thus, if we could prove that for a 2-interpolation family F, the associated function α from (13) is convex, as it is for 1-interpolations, then we would recover Santaló's inequality. This would be the case if we had a Brascamp-Lieb inequality with a factor $1/2$ on the right-hand side of (15) for every convex function $F : \mathbb{R}^n \to \mathbb{R}$. However, this is of course false in general. Recall that even for the Santaló inequaliy, some "center" must be fixed or some symmetry must be assumed. Therefore, a more reasonable question to ask, is whether α is convex when the initial data f is even. This guarantees that F_t is even for all $t \in [0, 1]$. However, it is again false in general that the Brascamp-Lieb inequality holds with factor $1/2$ in the right-hand side of (15) when F and u are even, as can be shown by taking a perturbation of the Gaussian measure. This suggests that the answer to the question could be negative in general. A reasonable conjecture, perhaps, is:

Conjecture 10. *Assume F_0 and F_1 are even, convex and 2-homogeneous (i.e. $F_i(x) = \lambda_i \|x\|_{K_i}^2$ for some centrally-symmetric convex bodies $K_i \subset \mathbb{R}^n$), properties that propagate along the interpolation. Then, the function α associated with the 2-interpolation family is convex.*

Here is a much more modest result:

Fact 11. *Assume that f is convex and even, and let F be a 2-interpolation family with $F_0 = f$ and $F_1 = \mathcal{L}f$, with the associated function α as in (13). Then, one has*

Figure 1: Conjecture 10 and its motivation in the source, PDF page 11.

2 Idea of the proof

For a smooth strictly convex body $K \subset \mathbb{R}^2$, put $F = \|\cdot\|_K^2/2$. Polar integration separates the Gaussian radial variable from the cone measure on ∂K. If a 2-homogeneous speed u has boundary value a, then the strengthened Brascamp–Lieb inequality needed in (3) becomes a Poincaré inequality on the centro-affine circle with spectral constant $4 = 2n$.

The Euclidean circle has exactly this constant, with equality for quadratic speeds. But the constant is not stable under nonellipsoidal perturbation. For the fourth Fourier perturbation $h = 1 + \varepsilon \cos 4\theta$, the quadratic-looking test $a = \cos 2\theta$ has Rayleigh deficit $8\pi\varepsilon(3 + 4\varepsilon)$. A small negative ε preserves strict convexity and makes the deficit negative. This proves negative curvature of α at one time. Analytic local existence for the interpolation PDE then converts the infinitesimal datum into a genuine family and hence a counterexample with two honest endpoints.

3 The explicit counterexample

Theorem 1. *There exist C^ω_+ unconditional centrally symmetric convex bodies $K_0, K_1 \subset \mathbb{R}^2$ and a real 2-interpolation family*

$$F_s(x) = \frac{1}{2}\|x\|_{K_s}^2, \qquad 0 \le s \le 1,$$

with endpoints F_0, F_1, such that the function in (2) is not convex. More precisely, the family may be chosen analytic on $(0, 1) \times (\mathbb{R}^2 \setminus \{0\})$ and satisfies $\alpha''(1/2) < 0$.

We split the proof into the geometric computation and the local-existence step.

Lemma 1 (Cone-coordinate second variation)**.** *Let $K \subset \mathbb{R}^2$ be C^∞_+ and centrally symmetric. Write its support function as $h(\theta)$ and its inverse Gauss parametrization as*

$$X(\theta) = h(\theta)n(\theta) + h'(\theta)t(\theta), \quad n = (\cos\theta, \sin\theta), \quad t = (-\sin\theta, \cos\theta).$$

Put $q = h + h''$, $F_* = \| \cdot \|_K^2/2$, and define the 2-homogeneous function u_* by

$$u_*(rX(\theta)) = r^2 a(\theta), \qquad r > 0.$$

If $\int_0^{2\pi} ahq\, d\theta = 0$, then the second variation prescribed by the interpolation equation at the Cauchy datum (F_*, u_*) is

$$\alpha''(0) = \frac{A - 4D}{M}, \quad A = \int_0^{2\pi} h^2(a')^2\, d\theta, \quad D = \int_0^{2\pi} a^2hq\, d\theta, \quad M = \int_0^{2\pi} hq\, d\theta. \tag{4}$$

Proof. The Jacobian of $x = rX(\theta)$ is

$$dx = r\det(X, X')\, dr\, d\theta = rhq\, dr\, d\theta.$$

Euler homogeneity and differentiation along ∂K give the orthogonal identities

$$\operatorname{Hess} F_*(X, X) = 1, \qquad \operatorname{Hess} F_*(X, X') = 0, \qquad \operatorname{Hess} F_*(X', X') = \frac{q}{h}.$$

Since $du_*(X) = 2a$ and $du_*(X') = a'$, it follows that

$$(\operatorname{Hess} F_*)^{-1}\nabla u_* \cdot \nabla u_* = r^2\left(4a^2 + \frac{h}{q}(a')^2\right). \tag{5}$$

The elementary radial integrals are

$$\int_0^\infty re^{-r^2/2}\, dr = 1, \quad \int_0^\infty r^3e^{-r^2/2}\, dr = 2, \quad \int_0^\infty r^5e^{-r^2/2}\, dr = 8.$$

Thus $\int e^{-F_*} = M$, the mean of u_* is zero, and

$$\operatorname{Var}_{\mu_{F_*}}(u_*) = \frac{8D}{M}, \qquad \int (\operatorname{Hess} F_*)^{-1}\nabla u_* \cdot \nabla u_*\, d\mu_{F_*} = \frac{2(4D + A)}{M}.$$

Substitution in (3) proves (4). □

Proof of Theorem 1. Take

$$h(\theta) = 1 + \varepsilon\cos 4\theta, \qquad a(\theta) = \cos 2\theta, \qquad \varepsilon = -\frac{1}{100}. \tag{6}$$

Then

$$h > 0, \qquad q = h + h'' = 1 - 15\varepsilon\cos 4\theta > 0,$$

so h is the support function of a C^ω_+ unconditional body K. Also $\int ahq = 0$ by Fourier orthogonality. Direct integration gives

$$M = \pi(2 - 15\varepsilon^2), \tag{7}$$

$$A = \pi(4 - 4\varepsilon + 2\varepsilon^2), \tag{8}$$

$$D = \pi\left(1 - 7\varepsilon - \frac{15}{2}\varepsilon^2\right). \tag{9}$$

Consequently,

$$A - 4D = 8\pi\varepsilon(3 + 4\varepsilon) = -\frac{148\pi}{625} < 0. \tag{10}$$

Lemma 1 therefore gives a negative formal second variation.

It remains to realize the Cauchy datum by an interpolation family. In ordinary polar coordinates $x = r(\cos\phi, \sin\phi)$, write $F(t, x) = r^2 f(t, \phi)$ and $v = f_t$. Equation (1) becomes

$$f_{tt} = \frac{2(2f + f_{\phi\phi})v^2 - 2f_\phi v v_\phi + f(v_\phi)^2}{2f(2f + f_{\phi\phi}) - (f_\phi)^2}. \tag{11}$$

The body in (6) and its Gauss parametrization are analytic. Moreover, the derivative of the Euclidean polar angle of $X(\theta)$ is $hq/|X|^2 > 0$, so normal angle and polar angle are related by an analytic diffeomorphism. Hence F_* and u_* correspond to analytic periodic Cauchy data $f(0, \phi)$ and $f_t(0, \phi)$. The denominator in (11) is the determinant of $\operatorname{Hess} F_*$ in the polar orthonormal frame and is strictly positive. The analytic Cauchy–Kowalevski theorem [2] therefore supplies a periodic analytic solution for $|t| < \tau_0$. Compactness of the circle and uniqueness on overlapping angular charts make τ_0 uniform.

Positivity and strict convexity persist after decreasing τ_0. The reduced ansatz makes 2-homogeneity automatic, while uniqueness in the analytic Cauchy problem preserves the coordinate reflection symmetries of the data. Hence every F_t is an even, unconditional, convex 2-homogeneous function and can be written $F_t = \|\cdot\|_{K_t}^2/2$ for a C_+^ω body K_t. Differentiation under the integral is justified uniformly by the quadratic upper and lower bounds on F_t. Equations (4) and (10) yield $\alpha''(0) < 0$.

Finally choose $0 < \tau < \tau_0$ and set $\widetilde{F}_s = F_{(2s-1)\tau}$ for $0 \le s \le 1$. Affine time changes preserve (1), and $\widetilde{\alpha}''(1/2) = 4\tau^2\alpha''(0) < 0$. Taking $K_0 = K_{-\tau}$ and $K_1 = K_\tau$ completes the counterexample. $\square$

4 Interpretation of smoothness at the origin

A globally C^2 function on $\mathbb{R}^n$ that is exactly 2-homogeneous is a quadratic form: its Hessian at the origin determines the function by scaling. Thus a literal reading of the source's preliminary "smooth function on $\mathbb{R}^n$" definition would reduce Conjecture 10 to ellipsoids, where α is affine. The conjecture explicitly writes its data as arbitrary norm squares, so its only nontrivial interpretation is the standard one used for smooth norms: C^∞ and strongly convex on $\mathbb{R}^n \setminus \{0\}$, continuous and convex at the origin, with the PDE holding off the origin (and hence almost everywhere). Theorem 1 lies exactly in this intended class.

5 Verification and novelty boundary

The algebra in (7)–(10) is independently checked by the included SymPy script. The sign has a large exact margin: the deficit is $-148\pi/625$, while $h \ge 99/100$ and $q \ge 17/20$. No numerical approximation is used in the proof.

A bounded search on 26 August 2026 covered the run indexes, the exact arXiv id and title, the phrases "real 2-interpolation," "Rochberg–Semmes interpolation," and "2-homogeneous Brascamp–Lieb," and citations visible from the source. It found no later paper stating a proof or counterexample to Conjecture 10. Recent centro-affine Poincaré work proves inequalities with constant n for unconditional bodies [3]; the conjecture here requires the stronger constant $2n$, and our planar calculation shows that this stronger constant fails. Independent prior observation remains possible, so the novelty label is candidate rather than certified.

The main human-review point is the standard analytic local-existence step for the periodic homogeneous equation (11). Everything before that step is an exact two-line Fourier computation, and the PDE is explicitly in noncharacteristic analytic normal form.

References

[1] D. Cordero-Erausquin and B. Klartag, *Interpolations, convexity and geometric inequalities*, in *Geometric Aspects of Functional Analysis*, Lecture Notes in Math. 2050, Springer, 2012, 151–168; arXiv:1109.3652.

[2] F. John, *Partial Differential Equations*, 4th ed., Applied Mathematical Sciences 1, Springer, 1982.

[3] Y. Hu and M. N. Ivaki, *Centro-affine Poincaré inequality: unconditional convex bodies*, arXiv:2607.20223.